Proof ledger

Repository source: proof/PROOF_LEDGER.md

Status labels:

- **Sourced**: stated and proved in an identified reference.
- **Hand-checked**: algebra reviewed in the project notes.
- **Exact-code checked**: a symbolic identity was verified over general symbolic parameters.
- **Numerically cross-checked**: deterministic quadrature/random tests agree.
- **Formal write-up drafted**: a line-by-line proof is in the repository, but has not received outside review.
- **Integrated manuscript drafted**: the claim has been merged into the LaTeX research note and compiled.
- **Hosted CI passed**: all configured GitHub Actions jobs completed successfully in a clean hosted environment.
- **Fresh-clone smoke-tested**: a newly cloned checkout ran all substantive checks and builds; generated-PDF byte differences, if any, are recorded separately.
- **Needs local rerun**: a matching check exists but should be rerun and preserved in the current repository state.
- **Needs external review**: no independent subject-matter expert has checked it.

ID	Claim	Current evidence	Status
N1	Support-function constraints and volume formula	Nishioka, Sections 2–4	Sourced
N2	Smooth approximation and volume continuity	Nishioka, Lemma 2; Martini–Montejano–Oliveros, Theorem 8.5.1; limiting argument in `BRIDGE_LEMMAS.md`	Sourced; formal write-up drafted
S1	Degree-3 coefficient is 1/22; tail coefficient is at most 1/58	Nishioka, Lemma 4 and Proposition 1; explicit derivation in `BRIDGE_LEMMAS.md`	Sourced; formal write-up drafted
S2	Tensor quadratic form is orthogonal across harmonic degrees	Polarized Bochner calculation in `BRIDGE_LEMMAS.md`, Lemmas 2.1–2.2	Hand-checked; formal write-up drafted; needs external review
T1	General cubic tensor is harmonic iff symmetric trace-free	`DEGREE3_TENSOR_IDENTITY.md`, Section 2; symbolic Laplacian check	Exact-code checked; formal write-up drafted; needs external review
T2	`||A_Y||_2^2 = (176*pi/7) tau`	Moment derivation in `DEGREE3_TENSOR_IDENTITY.md`, Section 5; general symbolic integration	Exact-code checked; formal write-up drafted; needs external review
T3	Exact formula for `||4 det A_Y||_2^2`	Quartic contraction derivation in `DEGREE3_TENSOR_IDENTITY.md`, Sections 6–8; general symbolic integration	Exact-code checked by two formulations; formal write-up drafted; needs external review

ID	Claim	Current evidence	Status
T4	Compact pointwise determinant formula	`DEGREE3_TENSOR_IDENTITY.md`, Sections 3–4; matrix formula compared modulo `x^2+y^2+z^2-1`	Exact-code checked; hand-checked; needs external review
T5	Rotational invariance and random convention checks	Deterministic sphere quadrature and random rotations	Numerically cross-checked independently in Python and R; both earlier runs pass
T6	Pairing counts for the six quartic moments	Hand count in `DEGREE3_TENSOR_IDENTITY.md`, Section 7; exhaustive exact pairing enumeration	Exact-count checks passed independently in Python and R
F1	Nuclear-norm identity and Cauchy–Schwarz step	`BRIDGE_LEMMAS.md`, Lemmas 4.1–4.2	Hand-checked; formal write-up drafted; needs external review
F2	Projection/duality step `||C||_2^2 = <A_h,C> <= (d/2) integral ||C||_*`	`BRIDGE_LEMMAS.md`, Corollary 2.3 and Lemmas 5.1–5.2	Hand-checked; formal write-up drafted; needs external review
A1	Final exact coefficient, decimal, and strict improvement	SymPy exact arithmetic; `BRIDGE_LEMMAS.md`, Theorem 6.2 and Corollary 6.3	Exact-code checked; hand-checked; integrated manuscript drafted
R1	Passage to nonsmooth bodies	Nishioka's approximation lemma; `BRIDGE_LEMMAS.md`, Theorem 7.1	Sourced; formal write-up drafted; needs external review
P1	Complete paper source compiles without unresolved references	`paper/degree3_spectral_refinement.tex`; `build_paper.sh`	Author-review revision integrated; runtime-tested in project environment
P2	PDF has no visible clipping, overlap, or broken glyphs	Ten release-candidate pages inspected at 180 dpi; PDF preflight passed	Render-checked
P3	Named-author understanding review of the proof architecture	Twelve-question structured review and resulting exposition changes	Completed; revised manuscript and supporting package approved for release-candidate preparation
C1	Hosted clean-environment checks for Python, R, the paper, metadata, and Markdown PDFs	`.github/workflows/verification.yml`; three green jobs confirmed for the tested commit on September 4, 2026	Hosted CI passed
C2	Fresh-clone reproduction of checks and builds	`CLEAN_CLONE_CHECK.md`; commits `96cea5727507` and `ff5118cc05ad`	Fresh-clone smoke-tested; all substantive stages passed; release-candidate run had two-byte differences in two generated documentation PDFs
D1	Public-facing verification and reproduction instructions	`VERIFICATION_STATUS.md`; `REPRODUCIBILITY.md`	Drafted and integrated

ID	Claim	Current evidence	Status
M1	Authorship, AI provenance, citation, licensing, version, date, and public URLs	`AI_ASSISTANCE.md`; `CITATION.cff`; `LICENSE`; `VERSION`; `RELEASE_DATE`; manuscript declaration; metadata and release preflights	Integrated for v0.1.0
RC1	Public v0.1.0 release candidate	Versioned paper, public README wording, release notes, URLs, date, tag-aware workflow, and release-candidate fresh-clone smoke test	Prepared; exact tagged commit requires three green hosted jobs on `main` and on the version tag before publication
L1	Exact constant/decimal not found in a targeted web search	Search terms and limitations recorded in `PROOF_AUDIT.md`	Preliminary only; not a systematic novelty review
E1	Full theorem has been independently reviewed	None yet	Needs external review

Current bottleneck

The internal proof chain, named-author review, integrated manuscript, public release metadata, hosted verification workflow, and fresh-clone smoke testing are complete. The release-candidate smoke test passed every substantive check and build; two generated documentation PDFs exhibited trivial binary variation, which is documented but is not a release blocker. The remaining internal gate is to obtain three green hosted jobs on the reviewed commit, tag that commit `v0.1.0`, and require the three-job tag workflow to pass before the GitHub release is published.

After the tag is public, independent subject-matter review is the principal scientific next step.

Release rule

Do not describe the candidate bound as peer reviewed, certified, independently verified, or accepted while `E1` remains open. Release `v0.1.0` may be labeled **public research draft seeking independent mathematical verification** after the reviewed tagged commit has passed the hosted `main` and tag workflows.